Design Group

Submittal #23-01.01 230001 - HVAC General Requirements

DG Architectural and Engineering Facility Solutions, PLLC
1075 Windward Ridge Pkwy, Suite 100
Alpharetta, Georgia 30005
Phone: (770) 667-6250
Fax: (770) 667-6251

Project: 2102482-CCBCU-McComb MS-New Sales Center
310 W Presley Blvd.
McComb, Mississippi 39648

Warehouse Equipment/Fans

RESPONSIBLE CONTRACTOR:	Fitts and Goodwin, Inc.	**RECEIVED FROM:**	Thomas M. Fitts, Jr. **(Fitts and Goodwin, Inc.)**
APPROVER:	Wil Caceres Kyle Couch Timothy Dwyer		
DATE CREATED:	02/13/2024	**STATUS:**	No Exceptions Taken
FINAL DUE DATE:	02/22/2024	**REVISION:**	01
TYPE:	Product Information	**SPEC SECTION:**	230001 - HVAC General Requirements
SHIPPING METHOD:			
DESCRIPTION:			

COPIES TO:
Thomas M. Fitts, Jr. **(Fitts and Goodwin, Inc.)** , Parker Fitts **(Fitts and Goodwin, Inc.)** , Emma Sievert **(Fitts and Goodwin, Inc.)** , Timothy Dwyer **(Design Group - ATL)** , Sonia Dalvi **(Design Group - DAL)** , Christopher Ciocci **(Design Group - ATL)**

ATTACHMENTS:
Submittal 002 R1 - Warehouse Equipment, Fans.pdf

COMMENTS:

Timothy Dwyer, Senior Project Manager **DATE: 2/19/2024**

RESUBMITTAL FOR:

COCA-COLA WAREHOUSE
MCCOMB, MS

MECHANICAL CONTRACTOR

AIR COMFORT SYSTEMS
301 W. PRESLEY BLVD.
MCCOMB, MS 39648
(601) 684- 2092

SUBMITTED BY:

STEVENS MECHANICAL SYSTEMS, INC.
110 LONE WOLF DRIVE
MADISON, MS 39110
(601) 855-7150

DATE:

FEBRUARY 12, 2024

PRODUCTS:

OUTDOOR UNIT
HEAT PUMP

Design Group

barrywehmiller bw betterworld

HVAC General Rev1 Submittal Review	
Project Name	CCBCU McComb MS New Sales Center
Project Number	2102482
Date	2/19/2024
Submittal Name	HVAC General Rev1
Submittal No.	23-01
Submittal Rev. No.	1

STAMP

DESIGN GROUP FACILITY SOLUTIONS, INC.

☑ No Exceptions Taken ☐ Revise and Resubmit
☐ Exceptions Taken as Noted ☐ Rejected

February 19th, 2024

Reviewed By CaceresW

Review is only for general conformance with the design concept of the project and compliance with the information given in the construction documents. The contractor is responsible for dimensions to be confirmed and coordinated at the job site, for information that pertains to the fabrication process and to the means, methods, techniques, sequences and procedures of construction. Review does not authorize changes to contract sum, time or completion dates.

Equipment Tag	Review Comments	Notes
General	Reviewed As Noted	For equipment and actuactors coordinate Electrical Requirements with electrical contractor prior to acquiring product. Coordinate equipment BMS interface with the Controls contractor.
Coordination	For Coordination Only	Coordinate equipments size, location, and clearance requirements will all other trades
SSAU/SSCU-1 & 2	No Exceptions	
EF-5	No Exceptions	
Roof Hood	No Exceptions	
Gas Unit Heaters	No Exceptions	

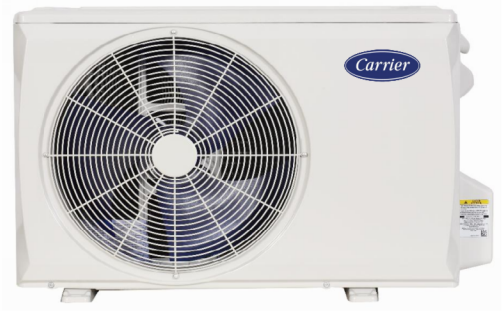
Outdoor Unit Single Zone Heat Pump Ductless System

38MARB Submittal Selection

Turn to the experts

Job Data: COCA-COLA **Location:** McCOMB MS **Carrier #:** 38MARBQ18

Tag #: SSCU-1,2 **Date:** 2/12/2024



NOTE: Images for illustration purposes only. Actual models may be slightly different.

Outdoor Model #	Size
38MARBQ12AA1	12K - 115V
38MARBQ06AA3	6K
38MARBQ09AA3	9K
38MARBQ12AA3	12K - 208/230V
38MARBQ18AA3	18K
38MARBQ24AA3	24K
38MARBQ30AA3	30K
38MARBQ36AA3	36K
38MARBQ36AB3	36K - AHU Only

Combination Selection	
Outdoor Unit	18K
Indoor Category	Cassette

STANDARD FEATURES
- Variable Speed (Inverter)
- Factory installed Base Pan Heater
- Factory installed Crankcase Heater
- Low Voltage Controls
- Auto-Restart function
- Condenser High Temp Protection
- Quiet operation
- Anti-corrosive fin coating

LIMITED WARRANTY*
- 10 year limited to original purchaser on compressor and parts upon timely registration, otherwise 5 years

*For residential applications.
See warranty for full details.

Specifications

Outdoor - Heat Pump											
System	Outdoor Size						18K				
Electrical	Voltage, Phase, Cycle	V/Ph/Hz					208/230-1-60				
	MCA	A.					16				
	Recommended Fuse Size	A.					20				
	MOCP - Fuse Rating	A.					25				
	Short Circuit Current Rating (SCCR)	kA					2.4				
Operating Range	Cooling Outdoor DB Min - Max	°F(°C)					-22~122 (-30~50)				
	Heating Outdoor DB Min - Max	°F(°C)					-22~86 (-30~30)				
Piping	Total Piping Length	ft (m)					98 (30)				
	Piping Lift*	ft (m)					65 (20)				
	Pipe Connection Size - Liquid	in (mm)					1/4 (6.35)				
	Pipe Connection Size - Suction	in (mm)					1/2 (12.7)				
Refrigerant	Refrigerant Type						R410A				
	Charge	lbs (kg)					4.08 (1.85)				
	Add'l Refrigerant (between Std & Max Piping Lngths)	Oz/ft (g/m)					0.161 (15)				
	Metering Device						EEV				
Outdoor Coil	Face Area	Sq. Ft.					5.9				
	No. Rows						2				
	Fins per inch						20				
	Circuits						6				
Compressor	Type						Rotary Inverter				
	Model						KTM240D43UKT				
	Oil Type						VG74				
	Oil Charge	Fl. Oz.					20.97				
	Rated Current	RLA					14.25				
Airflow & Sound	Airflow	CFM					1764.71				
	Sound Pressure	dB(A)					59				

* Condensing unit above or below indoor unit

Performance

	Outdoor Size		12K - 115V	6K	9K	12K - 208/230V	18K	24K	30K	36K	36K - AHU Only
High Wall	Indoor Model										
	Energy Star										
	Cooling Rated Capacity	Btu/h									
	Cooling Cap. Range Min - Max	Btu/h									
	SEER										
	EER										
	Heating Rated Capacity (47°F)	Btu/h									
	Heating Rated Capacity (17°F)	Btu/h									
	Heating Maximum Capacity (17°F	Btu/h									
	Heating Maximum Capacity (5°F)	Btu/h									
	Heating Cap. Range Min - Max	Btu/h									
	HSPF										
	COP (47°F)	W/W									
	COP (17°F)	W/W									
	COP (5°F)	W/W									

Copyright 2022 CAC/BDP • Atlanta, GA, 30339

SUBJECT TO CHANGE WITHOUT NOTICE

Catalog No.: 38MARB-02SB

Replaces: 38MARB-01SB

Edition Date: 07/2022

	Outdoor Size		12K - 115V	6K	9K	12K - 208/230V	18K	24K	30K	36K	36K - AHU Only
Cassette	Indoor Model						18K				
	Energy Star						YES				
	Cooling System Tons						1.33				
	Cooling Rated Capacity	Btu/h					16000				
	Cooling Cap. Range Min - Max	Btu/h					5,600~17,000				
	SEER						20				
	EER						12.5				
	Heating Rated Capacity (47°F)	Btu/h					18000				
	Heating Rated Capacity (17°F)	Btu/h					11000				
	Heating Maximum Capacity (17°F	Btu/h					21000				
	Heating Maximum Capacity (5°F)	Btu/h					17000				
	Heating Cap. Range Min - Max	Btu/h					8,700~21,000				
	HSPF						10.3				
	COP (47°F)	W/W					2.88				
	COP (17°F)	W/W					1.8				
	COP (5°F)	W/W					1.8				
Ducted	Indoor Model										
	Energy Star										
	Cooling System Tons										
	Cooling Rated Capacity	Btu/h									
	Cooling Cap. Range Min - Max	Btu/h									
	SEER										
	EER										
	Heating Rated Capacity (47°F)	Btu/h									
	Heating Rated Capacity (17°F)	Btu/h									
	Heating Maximum Capacity (17°F	Btu/h									
	Heating Maximum Capacity (5°F)	Btu/h									
	Heating Cap. Range Min - Max	Btu/h									
	HSPF										
	COP (47°F)	W/W									
	COP (17°F)	W/W									
	COP (5°F)	W/W									
Mix Match Ducted	Indoor Model										
	Energy Star										
	Cooling System Tons										
	Cooling Rated Capacity	Btu/h									
	Cooling Cap. Range Min - Max	Btu/h									
	SEER										
	EER										
	Heating Rated Capacity (47°F)	Btu/h									
	Heating Rated Capacity (17°F)	Btu/h									
	Heating Maximum Capacity (17°F	Btu/h									
	Heating Maximum Capacity (5°F)	Btu/h									
	Heating Cap. Range Min - Max	Btu/h									
	HSPF										
	COP (47°F)	W/W									
	COP (17°F)	W/W									
	COP (5°F)	W/W									
Console	Indoor Model										
	Energy Star										
	Cooling System Tons										
	Cooling Rated Capacity	Btu/h									
	Cooling Cap. Range Min - Max	Btu/h									
	SEER										
	EER										
	Heating Rated Capacity (47°F)	Btu/h									
	Heating Rated Capacity (17°F)	Btu/h									
	Heating Maximum Capacity (17°F	Btu/h									
	Heating Maximum Capacity (5°F)	Btu/h									
	Heating Cap. Range Min - Max	Btu/h									
	HSPF										
	COP (47°F)	W/W									
	COP (17°F)	W/W									
	COP (5°F)	W/W									
Air Handler	Indoor Model										
	Energy Star										
	Cooling System Tons										
	Cooling Rated Capacity	Btu/h									
	Cooling Cap. Range Min - Max	Btu/h									
	SEER										
	EER										
	Heating Rated Capacity (47°F)	Btu/h									
	Heating Rated Capacity (17°F)	Btu/h									
	Heating Maximum Capacity (17°F)	Btu/h									
	Heating Maximum Capacity (5°F)	Btu/h									
	Heating Cap. Range Min - Max	Btu/h									
	HSPF										
	COP (47°F)	W/W									
	COP (17°F)	W/W									
	COP (5°F)	W/W									

Copyright 2022 CAC/BDP • Atlanta, GA, 30339

Catalog No.: 38MARB-02SB

SUBJECT TO CHANGE WITHOUT NOTICE

Replaces: 38MARB-01SB

Outdoor Size		12K - 115V	6K	9K	12K - 208/230V	18K	24K	30K	36K	36K - AHU Only

Edition Date: 07/2022

FB4C Fan Coil	Indoor Model										
	Indoor Model 2										
	Indoor Model Capacity										
	Energy Star										
	Cooling System Tons										
	Cooling Rated Capacity	Btu/h									
	Cooling Cap. Range Min - Max	Btu/h									
	SEER										
	EER										
	Heating Rated Capacity (47°F)	Btu/h									
	Heating Rated Capacity (17°F)	Btu/h									
	Heating Maximum Capacity (5°F)	Btu/h									
	Heating Cap. Range Min - Max	Btu/h									
	HSPF										
	COP (47°F)	W/W									
	COP (17°F)	W/W									
	COP (5°F)	W/W									

FMA4X Fan Coil	Indoor Model										
	Indoor Model Capacity										
	Energy Star										
	Cooling System Tons										
	Cooling Rated Capacity	Btu/h									
	Cooling Cap. Range Min - Max	Btu/h									
	SEER										
	EER										
	Heating Rated Capacity (47°F)	Btu/h									
	Heating Rated Capacity (17°F)	Btu/h									
	Heating Maximum Capacity (5°F)	Btu/h									
	Heating Cap. Range Min - Max	Btu/h									
	HSPF										
	COP (47°F)	W/W									
	COP (17°F)	W/W									
	COP (5°F)	W/W									

FX4D Fan Coil	Indoor Model										
	Indoor Model Capacity										
	Energy Star										
	Cooling System Tons										
	Cooling Rated Capacity	Btu/h									
	Cooling Cap. Range Min - Max	Btu/h									
	SEER										
	EER										
	Heating Rated Capacity (47°F)	Btu/h									
	Heating Rated Capacity (17°F)	Btu/h									
	Heating Maximum Capacity (5°F)	Btu/h									
	Heating Cap. Range Min - Max	Btu/h									
	HSPF										
	COP (47°F)	W/W									
	COP (17°F)	W/W									
	COP (5°F)	W/W									

FV4C Fan Coil	Indoor Model										
	Indoor Model Capacity										
	Energy Star										
	Cooling System Tons										
	Cooling Rated Capacity	Btu/h									
	Cooling Cap. Range Min - Max	Btu/h									
	SEER										
	EER										
	Heating Rated Capacity (47°F)	Btu/h									
	Heating Rated Capacity (17°F)	Btu/h									
	Heating Maximum Capacity (5°F)	Btu/h									
	Heating Cap. Range Min - Max	Btu/h									
	HSPF										
	COP (47°F)	W/W									
	COP (17°F)	W/W									
	COP (5°F)	W/W									

FMC-FMU Fan Coil	Indoor Model										
	Indoor Model Capacity										
	Energy Star										
	Cooling System Tons										
	Cooling Rated Capacity	Btu/h									
	Cooling Cap. Range Min - Max	Btu/h									
	SEER										
	EER										
	Heating Rated Capacity (47°F)	Btu/h									
	Heating Rated Capacity (17°F)	Btu/h									
	Heating Maximum Capacity (5°F)	Btu/h									
	Heating Cap. Range Min - Max	Btu/h									
	HSPF										
	COP (47°F)	W/W									
	COP (17°F)	W/W									
	COP (5°F)	W/W									

Copyright 2022 CAC/BDP • Atlanta, GA, 30339

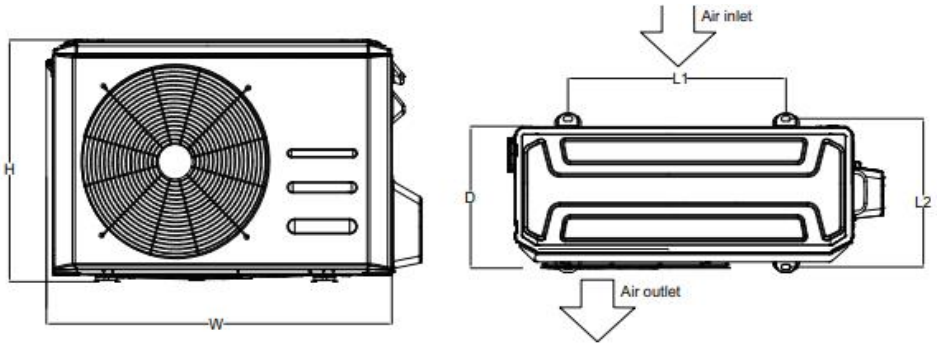
Catalog No.: 38MARB-02SB

SUBJECT TO CHANGE WITHOUT NOTICE

Replaces: 38MARB-01SB

Dimensions

Edition Date: 07/2022



Size						18K				
Width (W)	in (mm)					35.04 (890)				
Height (H)	in (mm)					26.50 (673)				
Depth (D)	in (mm)					13.46 (342)				
L1	in (mm)					26.1 (663)				
L2	in (mm)					13.65 (346.674)				
Operating Weight	lbs (kg)					100.97 (45.8)				
Shipping Width	in (mm)					39.17 (995)				
Shipping Height	in (mm)					29.13 (740)				
Shipping Depth	in (mm)					15.67 (398)				
Shipping Weight	lbs (kg)					108.03 (49)				

Copyright 2022 CAC/BDP • Atlanta, GA, 30339

SUBJECT TO CHANGE WITHOUT NOTICE

Catalog No.: 38MARB-02SB

Replaces: 38MARB-01SB

Edition Date: 07/2022

Cassette Heat Pump Ductless System

Indoor Model: 40MBCQ18---3

Submittal Data

Job Data:	COCA COLA	Location:	McComb MS
Buyer:	Air Comfort	Buyer P.O. #:	
Carrier #:	40MBCQ18		
Unit Number:	SSAU 1,2	Model Number:	40MBCQ18
Performance Data Certified By:		Date:	02-12-24





STANDARD FEATURES
Modes: Cool, Heat, Dry, Fan, Auto
Louver Control (Fixed or Swing)
Outside Air Intake
Built-in condensate Lift pump
Follow Me (senses temperature at handheld remote)
Heating Setback (46° F Heating Mode)
Quiet indoor operation
Anti-corrosive fin coating

Controls:
Wireless Remote Controller included with indoor unit
Optional Wired Remote Controller (Timer Function)
Optional Wired Remote Controller KSACN0501AAA (7 Day programmable)

LIMITED WARRANTY*
• 10 year limited to original purchaser on compressor and parts upon timely registration, otherwise 5 years
*For residential applications. See warranty for full details.

System	Size		18
	Indoor Model		40MBCQ18---3
Electrical	Voltage, Phase, Cycle	V/Ph/Hz	208/230-1-60
	Power Supply	Indoor unit powered by outdoor unit	
	MCA	A.	0.2
Controls	Wireless Remote Controller (°F/°C Convertible)		Standard
	Wired Remote Controller (°F/°C Convertible)		Optional
Operating Range	Cooling Indoor DB Min -Max	°F(°C)	63~90 (17~32)
	Heating Indoor DB Min -Max	°F(°C)	32~86 (0~30)
Piping	Pipe Connection Size - Liquid	in (mm)	1/4 (6.35)
	Pipe Connection Size - Suction	in (mm)	1/2 (12.7)

Indoor Coil	Face Area	Sq. Ft.	3.10
	No. Rows		2
	Fins per inch		19
	Circuits		4
Indoor	Body Unit Width	in (mm)	22.44 (570)
	Body Unit Height	in (mm)	10.24 (260)
	Body Unit Depth	in (mm)	22.44 (570)
	Body Net Weight	lbs (kg)	39.68 (18)
	Panel Unit Width	in (mm)	25.47 (647)
	Panel Unit Height	in (mm)	1.97 (50)
	Panel Unit Depth	in (mm)	25.47 (647)
	Panel Net Weight	lbs (kg)	5.51 (2.5)
	Number of Fan Speeds		3
	Airflow (low to high)	CFM	290/350/420
	Sound Pressure (lowest to highest)	dB(A)	46/48/50
	Air throw Data	ft (m)	30 (9)
	Moisture removal	Pint/h (L/h)	4.26 (2.02)
	Field Drain Pipe Size O.D.	in (mm)	1 (25.4)

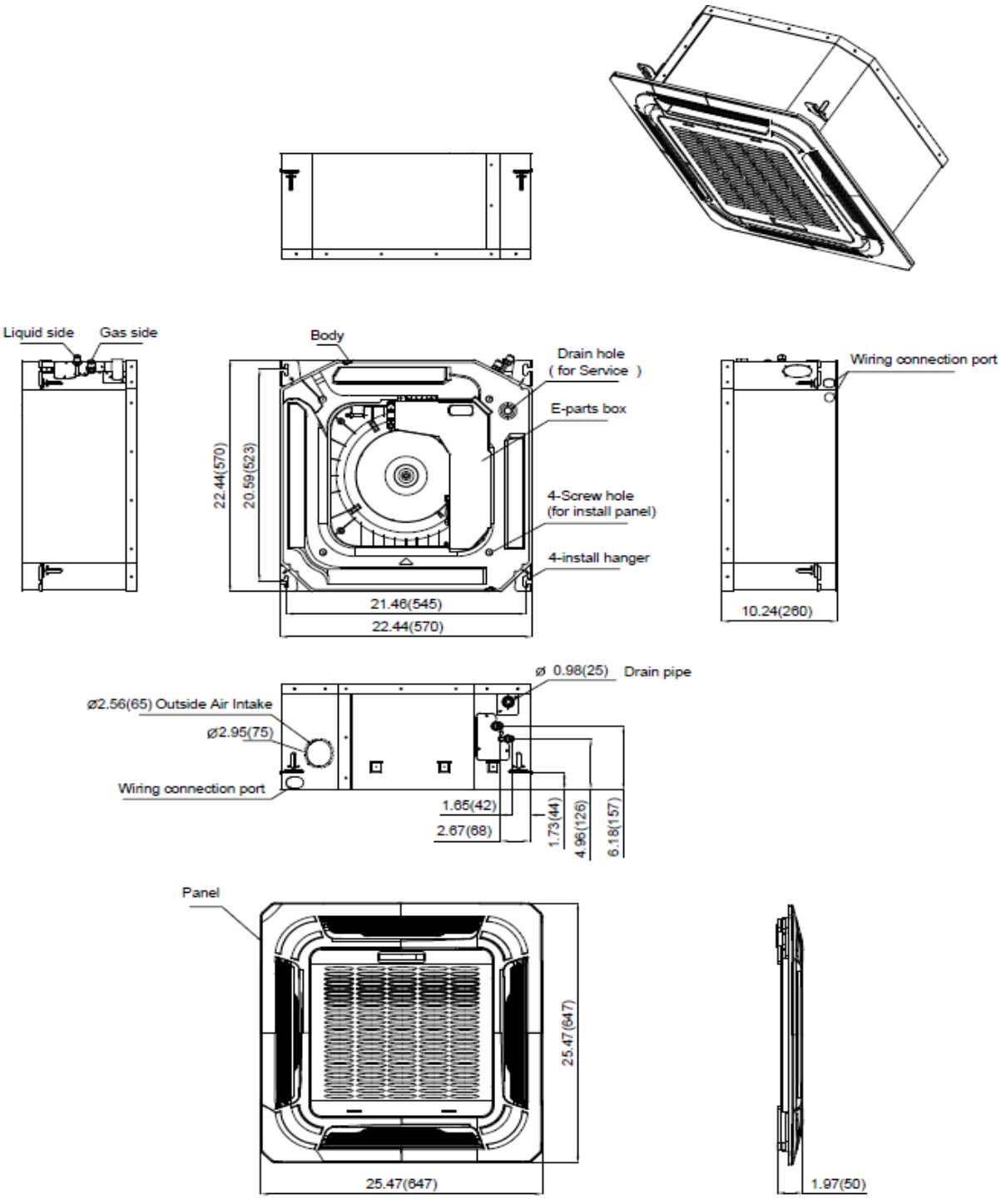
For Compatibility See Product Data
Performance may vary based on the outdoor unit matched to. See compatible outdoor units pages for Performance Data.

Copyright 2018 CAC/BDP ● 7310 W. Morris St. Indianapolis, IN 46231
Edition Date: 03/2018
Catalog No: 40MBCQ-18-3-02SB

Accessories

KSACN0101AAA	Wired Remote Control with Timer Function
KSACN0501AAA	Wired Remote Control 7 day Programmable
KSAIF0401AAA	Wi-Fi™ Kit
KSAIC0101230	24V Interface Kit 208/230V
40MBCQ01XXX3	Grille/Ceiling panel
53DS-900---008	Insulated 25' Line Set - 1/4" x 1/2"

Construction View



RESUBMITTAL FOR:

COCA-COLA WAREHOUSE
MCCOMB, MS

MECHANICAL CONTRACTOR

AIR COMFORT SYSTEMS
301 W. PRESLEY BLVD.
MCCOMB, MS 39648
(601) 684- 2092

SUBMITTED BY:

STEVENS MECHANICAL SYSTEMS, INC.
110 LONE WOLF DRIVE
MADISON, MS 39110
(601) 855-7150

DATE:

FEBRUARY 12, 2024

PRODUCTS:

EXHAUST FANS
ROOF CAP W/ CURB
GAS UNIT HEATERS

STEVENS MECHANICAL SYSTEMS INC.
110 LONE WOLF DR. MADISON, MS 39110
PH: 601.855.7150 FAX: 601.855.7390
STEVENS@STEVENSMECHANICAL.COM

COCA-COLA WAREHOUSE
MCCOMB, MS

AIR COMFORT SYSTEMS

EXHAUST FANS

"SEE ATTACHED"

Design Group

barrywehmiller bw betterworld

HVAC General Rev1 Submittal Review	
Project Name	CCBCU McComb MS New Sales Center
Project Number	2102482
Date	2/19/2024
Submittal Name	HVAC General Rev1
Submittal No.	23-01
Submittal Rev. No.	1

STAMP

DESIGN GROUP FACILITY SOLUTIONS, INC.
☑ No Exceptions Taken ☐ Revise and Resubmit
☐ Exceptions Taken as Noted ☐ Rejected
February 19th, 2024
Reviewed By CaceresW
Review is only for general conformance with the design concept of the project and compliance with the information given in the construction documents. The contractor is responsible for dimensions to be confirmed and coordinated at the job site, for information that pertains to the fabrication process and to the means, methods, techniques, sequences and procedures of construction. Review does not authorize changes to contract sum, time or completion dates.

Equipment Tag	Review Comments	Notes
General	Reviewed As Noted	For equipment and actuactors coordinate Electrical Requirements with electrical contractor prior to acquiring product. Coordinate equipment BMS interface with the Controls contractor.
Coordination	For Coordination Only	Coordinate equipments size, location, and clearance requirements will all other trades
SSAU/SSCU-1 & 2	No Exceptions	
EF-5	No Exceptions	
Roof Hood	No Exceptions	
Gas Unit Heaters	No Exceptions	

SUBMITTAL

Job Name: COCA-COLA WAREHOUSE

Job Site: MCCOMB, MS

Date: FEBRUARY 12, 2024

Submitted By: SMSI

GREENHECK
Building Value in Air.

P.O. Box 410 Schofield, WI 54476 (715) 359-6171 FAX (715) 355-2399 www.greenheck.com

Model: CSP-A700-VG

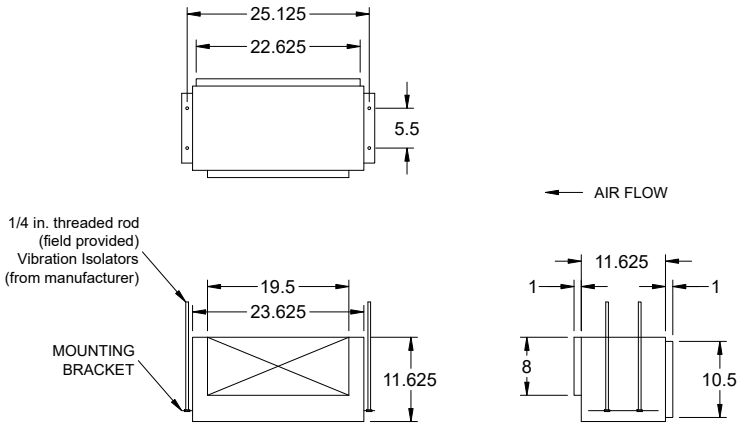
Inline Cabinet Fan

Standard Construction Features:

- Galvanized steel housing - Sound absorbing insulation - Outlet duct collar with integral spring loaded back draft damper - Forward curved wheel - Plug type disconnect - Adjustable mounting brackets - Square duct connection - Dual galvanized steel scroll design - Field rotatable discharge

Selected Options & Accessories:

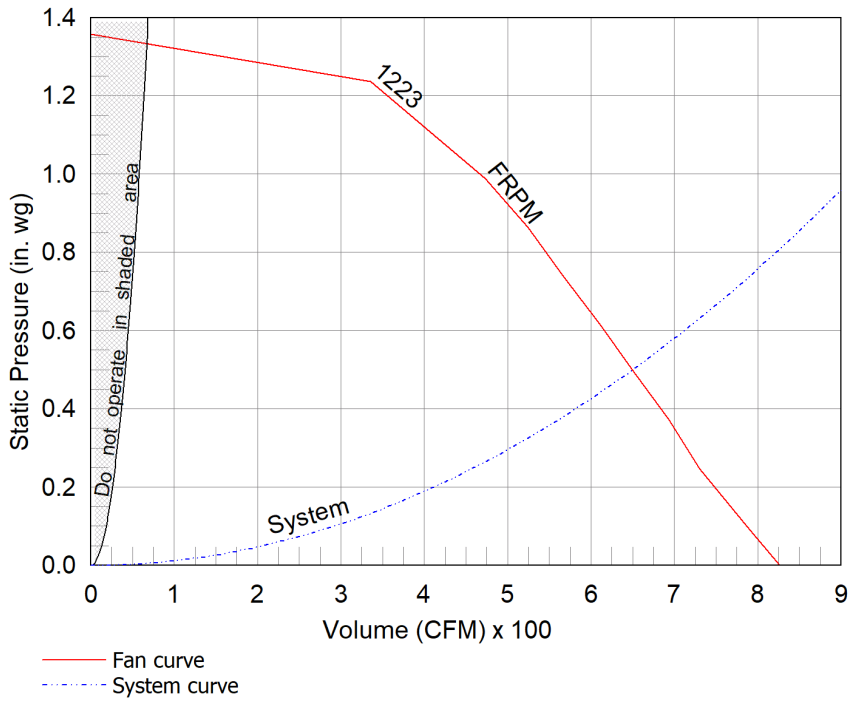
Motor rated for continuous use
Motor - Vari-Green EC motor w/Mounted Potentiometer Dial
UL/cUL 507 Listed - Electric Fan
Isolation Kit, (PN: VI KIT-SP/CSP), Shipped Loose
Adjustable easy installation mounting bracket
Polypropylene Wheel Material



Dimensional	
Quantity	1
Weight w/o Acc's (lb)	39
Weight w/ Acc's (lb)	40

Performance	
Requested Volume (CFM)	650
Actual Volume (CFM)	650
Total External SP (in. wg)	0.5
Fan RPM	1223
* FLA (A)	4.1
Elevation (ft)	331
Airstream Temp.(F)	70
Air Density (lb/ft3)	0.074
Sones	1.4

Motor	
Motor Mounted	Yes
** Input Watts (W)	142
Voltage/Cycle/Phase	115/60/1
Enclosure	TENV
Control Input Voltage	8



Notes:

All dimensions shown are in units of in.
*FLA is approximate and will vary slightly with the motor.
Sones - calculated using AMCA 301 at 5 ft
** Wattage is approximate and may vary between motors.

STEVENS MECHANICAL SYSTEMS INC.
110 LONE WOLF DR. MADISON, MS 39110
PH: 601.855.7150 FAX: 601.855.7390
STEVENS@STEVENSMECHANICAL.COM

COCA-COLA WAREHOUSE
MCCOMB, MS

AIR COMFORT SYSTEMS

ROOF CAP W/ CURB

"SEE ATTACHED"

Design Group

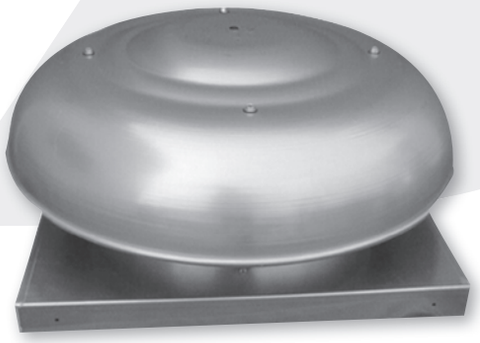
barrywehmiller bw betterworld

HVAC General Rev1 Submittal Review	
Project Name	CCBCU McComb MS New Sales Center
Project Number	2102482
Date	2/19/2024
Submittal Name	HVAC General Rev1
Submittal No.	23-01
Submittal Rev. No.	1

STAMP

DESIGN GROUP FACILITY SOLUTIONS, INC.
☑ No Exceptions Taken ☐ Revise and Resubmit
☐ Exceptions Taken as Noted ☐ Rejected
February 19th, 2024
Reviewed By CaceresW
Review is only for general conformance with the design concept of the project and compliance with the information given in the construction documents. The contractor is responsible for dimensions to be confirmed and coordinated at the job site, for information that pertains to the fabrication process and to the means, methods, techniques, sequences and procedures of construction. Review does not authorize changes to contract sum, time or completion dates.

Equipment Tag	Review Comments	Notes
General	Reviewed As Noted	For equipment and actuactors coordinate Electrical Requirements with electrical contractor prior to acquiring product. Coordinate equipment BMS interface with the Controls contractor.
Coordination	For Coordination Only	Coordinate equipments size, location, and clearance requirements will all other trades
SSAU/SSCU-1 & 2	No Exceptions	
EF-5	No Exceptions	
Roof Hood	No Exceptions	
Gas Unit Heaters	No Exceptions	



WEATHER CAPS

Models: WCF & WCC
Rooftop Exhaust or Supply Ventilators
Flat Base and Curb Mount Base

MOVING YOUR WAY

› WCF & WCC Exhaust Performance Data

Model	Exhaust CFM Pressure Drop ("WC)								
	0.025	0.050	0.100	0.125	0.150	0.200	0.250	0.300	0.350
WCF/WCC06	93	131	185	207	227	262	293	320	346
WCF/WCC08	221	313	442	494	541	625	699	766	827
WCF/WCC10	297	420	594	664	727	840	939	1029	1111
WCF/WCC12	493	697	986	1102	1208	1394	1559	1708	1845
WCF/WCC14	604	853	1207	1349	1478	1707	1908	2091	2258
WCF/WCC16	746	1055	1492	1668	1827	2110	2359	2584	2791
WCF/WCC18	919	1300	1838	2055	2251	2599	2906	3184	3439
WCF/WCC20	1127	1593	2253	2519	2759	3186	3562	3902	4215
WCC24	1471	2080	2941	3288	3602	4159	4650	5094	5502
WCC30	1768	2500	3535	3952	4329	4999	5589	6123	6613
WCC36	2730	3860	5459	6103	6686	7720	8631	9455	10213
WCC42	3468	4904	6936	7755	8495	9809	10967	12014	12976
WCC48	5363	7584	10725	11991	13135	15167	16958	18576	20065

› WCF & WCC Intake Performance Data

Model	Intake CFM Pressure Drop ("WC)								
	0.025	0.050	0.100	0.125	0.150	0.200	0.250	0.300	0.350
WCF/WCC06	64	91	128	143	157	181	202	222	239
WCF/WCC08	148	209	296	331	363	419	468	513	554
WCF/WCC10	202	286	404	452	495	571	639	700	756
WCF/WCC12	330.5	467	661	739	810	935	1045	1145	1237
WCF/WCC14	440.5	623	881	985	1079	1246	1393	1526	1648
WCF/WCC16	552	781	1104	1234	1352	1561	1746	1912	2065
WCF/WCC18	707.5	1001	1415	1582	1733	2001	2237	2451	2647
WCF/WCC20	856	1211	1712	1914	2097	2421	2707	2965	3203
WCC24	1191	1684	2382	2663	2917	3369	3766	4126	4456
WCC30	1537.5	2174	3075	3438	3766	4349	4862	5326	5753
WCC36	2402	3397	4804	5371	5884	6794	7596	8321	8987
WCC42	3052	4316	6104	6824	7476	8632	9651	10572	11420
WCC48	4612	6522	9224	10313	11297	13045	14584	15976	17257

› Product Configuration

› Model
WCF = Weather Cap Flat
WCC = Weather Cap Curb

› Unit Size
06 08 10 12 14 16 18
20 24 30 36 42 48

› Application E/S
E = Exhaust
S = Supply

› Paint/Coating
0 = None
F = Epoxy Powder Coat*
G = Epoxy Powder Coat w/ UV*
H = Hi-Temp Powder Coat*
J = Non-stick Powder Coat*
K = Phenolic Powder Coat*
L = Phenolic Powder Coat w/ UV*
N = Polyester Powder Coat
X = Special
** Not available with choice of color.*

› Color
0 = None
50 = Chrome Green
53 = Williamsburg Blue
55 = Pale Green
56 = Dove Gray
61 = White
63 = Oxford Beige
65 = Dover White
66 = Desert Tan
70 = Black
73 = Smoke Gray
77 = Brick Red
79 = Peppercorn
81 = Pale Brown
83 = Chocolate Brown
85 = Timeless Bronze
94 = Charcoal
X = Special

› Roof Slope
0 = None
S = Single Slope
D = Double Slope

› Roof Curb
0 = None
B = UCG12
D = UCA8
F = SFA12
H = SCG16
K = UCA18
M = SA16
P = SFG18
R = SRA16
T = UA12
V = UG18
Y = URA18
10 = SFA8
12 = USCA
4 = UVA18
A = UCG8
C = UCG18
E = UCA12
G = SFA18
J = SCA16
L = UG12
N = SFG12
Q = SG16
S = SRG16
U = UA18
W = URA12
1 = URG12
11 = USCG
2 = URG18
5 = UVG18

› Damper
0 = None
BDD = Gravity Backdraft
CBD = Counter Balance Relief
MPBD1 = Motor Operated 115V
MPBD2 = Motor Operated 230V
MPBD4 = Motor Operated 450V

› Metal Liner
0 = None
L = Liner

› Damper Holding Plate
0 = None
P = Damper Holding Plate

› Neoprene Gasket
0 = None
G = Gasket

› No Wooden Nailer
0 = None
N = No Wooden Nailer

› Curb Paint/Coating
0 = None
B = Air Dried Epoxy
Q = Enamel

› Screen
0 = None
S = Insect Screen

©PENNBARRY 2008

STEVENS MECHANICAL SYSTEMS INC.
110 LONE WOLF DR. MADISON, MS 39110
PH: 601.855.7150 FAX: 601.855.7390
STEVENS@STEVENSMECHANICAL.COM

COCA-COLA WAREHOUSE
MCCOMB, MS

AIR COMFORT SYSTEMS

GAS UNIT HEATERS

"SEE ATTACHED"

Design Group

barrywehmiller bw betterworld

HVAC General Rev1 Submittal Review	
Project Name	CCBCU McComb MS New Sales Center
Project Number	2102482
Date	2/19/2024
Submittal Name	HVAC General Rev1
Submittal No.	23-01
Submittal Rev. No.	1

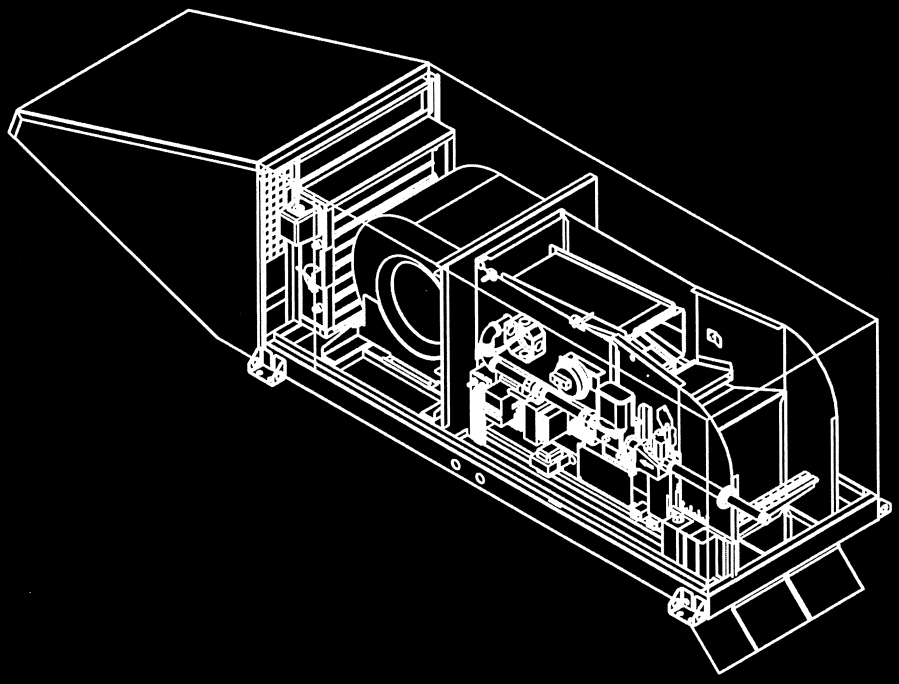
STAMP

DESIGN GROUP FACILITY SOLUTIONS, INC.
☑ No Exceptions Taken ☐ Revise and Resubmit
☐ Exceptions Taken as Noted ☐ Rejected
February 19th, 2024
Reviewed By CaceresW
Review is only for general conformance with the design concept of the project and compliance with the information given in the construction documents. The contractor is responsible for dimensions to be confirmed and coordinated at the job site, for information that pertains to the fabrication process and to the means, methods, techniques, sequences and procedures of construction. Review does not authorize changes to contract sum, time or completion dates.

Equipment Tag	Review Comments	Notes
General	Reviewed As Noted	For equipment and actuactors coordinate Electrical Requirements with electrical contractor prior to acquiring product. Coordinate equipment BMS interface with the Controls contractor.
Coordination	For Coordination Only	Coordinate equipments size, location, and clearance requirements will all other trades
SSAU/SSCU-1 & 2	No Exceptions	
EF-5	No Exceptions	
Roof Hood	No Exceptions	
Gas Unit Heaters	No Exceptions	

CAMBRIDGE SA-SERIES

BLOW THRU - DIRECT GAS-FIRED SPACE HEATERS ENGINEERING SPECIFICATIONS MANUAL



250 to 350 MBH Capacities
Complete Heater Packages & Options

CAMBRIDGE®
AIR SOLUTIONS
Enriching Lives

760 Long Road Crossing Dr., Chesterfield MO 63005 • (636) 532-2233 • (800) 899-1989 • Fax (636) 530-6133 • www.cambridgeair.com

ASHRAE 90.1 COMPLIANT

Copyright © 2023 Cambridge Air Solutions. All Rights Reserved.

SA-ES4-0523

Being the best at saving energy, reducing operating costs and improving indoor air quality has made Cambridge Air Solutions the preferred choice for heating warehouses, manufacturing plants, automobile service areas, aircraft hangars, indoor recreational facilities, and other commercial/industrial buildings. Other manufacturers now describe their heaters as being "Just Like Cambridge®." Don't believe it! Only Cambridge offers Blow-Thru® Space Heating Technology that outperforms every other direct gas-fired heater on the market.

Application

- **Flexible Application** - Cambridge® SA-Series Heaters can be used as a perimeter heating system or rotating air system; and as an air-neutralization or air-pressurization system. When properly applied, Cambridge Blow-Thru® Space Heaters will heat the least amount of fresh air required to address the facility's air infiltration, make-up air and space heating needs.
- **High Induction Ratio** – The high velocity discharge air induces large volumes of room air into the discharge stream, at a ratio exceeding 10:1. This process mixes the high temperature discharge air with room air, providing large volumes of fresh, warm air flowing throughout the building, thus eliminating higher ceiling temperature and uncomfortable drafts.

Certification

- **Tested and Certified by CSA International** - Cambridge is certified to the stringent requirements of ANSI Standard Z83.4/CSA3.7 for non-recirculating air heaters. SA-Series Heaters are approved for use in both the U.S. and Canada.
- **ASHRAE 90.1 Compliant** - Properly configured energy efficient Cambridge® Heaters comply with the latest requirements of ASHRAE Standard 90.1. This is a requirement for LEED Certified Green Buildings and a growing number of local, state and federal (DOE) building codes.
- **Safe** – Non-recirculating Cambridge® Heaters have less than 5.0 ppm carbon monoxide and 0.5 ppm nitrogen dioxide at all firing rates. SA-Series Heaters use 100%fresh outside air to improve indoor air quality.
- **Max Performance** - Cambridge SA-Series heaters are certified for 160°F maximum discharge temperature, the highest allowed by the standard. This higher discharge temperature rating is crucial for space heating applications where the cost of energy matters.

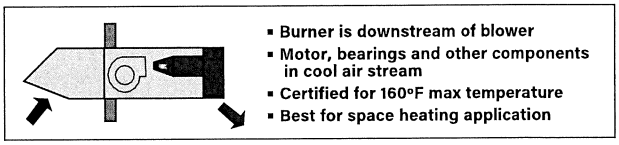
Proprietary Cambridge Burner

Cambridge designs and manufactures its own proprietary high efficiency, stainless steel burners. You can't get the high performance and extended life of a Cambridge burner unless you have a Cambridge® Heater.

Blow-Thru® Space Heater Design

The SA-Series Heater uses Blow-Thru® Space Heating Technology, where the proprietary Cambridge burner is located downstream of the blower. This translates into a smaller, more energy efficient design for space heating applications compared to a conventional draw-thru make-up air heater. Cambridge® SA-Series Heaters use a lower horsepower motor and, most important of all, use less outside air to heat a building. This is because of its higher Btu/cfm ratio and unique certified 160°F maximum rating for discharge temperature.

Cambridge Blow-Thru® Heater



Save Energy

High temperature rise, Blow-Thru® Space Heaters are 40% to 70% more energy efficient than all other types of indirect and direct gas-fired systems used to heat/ventilate large commercial and industrial buildings. This has been documented by computer energy modeling and a growing list of over 300 on-site building studies.

Other Industrial Heating Systems	Energy Savings with Cambridge® Space Heaters*
Boilers	40% to 70%
Unit Heaters	30% to 50%
Air Turnover Systems	25% to 70%
Infrared (Radiant)	15% to 40%
Make-Up Air (MUA)	20% to 50%
Recirculation (80/20 - pressurization)	20% to 50%

* Some building studies show more energy savings than listed above

Reduce Carbon Footprint

Saving energy means a corresponding reduction in CO2 emissions to reduce your building's carbon footprint. Cambridge heaters are recommended for LEED/Green projects.

Lower Installation Costs

Smaller, lightweight, pre-piped and pre-wired SA-Series Heaters are available with three mounting options including the popular thru-wall design for easier installation. Every heater is tested as a fully assembled unit to minimize initial start-up problems.

Reliability

Genuine Cambridge® Heaters require little maintenance and are built to last. The Blow-Thru® design means the motor and other critical components are located in the cool air stream for extended life. Patented Low Fire Start technology increases reliability and saves energy. The industry's best warranty package includes 2 years on the heater and 5 years on the stainless Cambridge burner.

Specifications

SA-Series HTHV Blow-Thru® Space Heater

General: The high efficiency, SA-Series Direct Gas-Fired Heater shall be manufactured, assembled and factory tested to assure proper alignment of assemblies and performance of controls and other components. Each heater must be design certified by CSA International to be in compliance with ANSI Standard Z83.4 • CSA 3.7 for Non-Recirculating Direct Gas-Fired Industrial Air Heaters and be labeled ASHRAE 90.1 compliant. Controls and gas train comply with the requirements of Factory Mutual (FM) and XL Insurance Global Asset Protection Services (XL GAPS). **Each heater shall be capable of achieving a 160°F discharge temperature in accordance with the terms of the certification.** The standard heater mounting package, including all specified components, shall be ❑ thru wall ❑ roof top ❑ under roof ❑ outdoor vertical ❑ indoor vertical.

Construction: Each **Blow-Thru®** space heater shall be constructed with the **burner section located at the blower discharge**, isolating the blower, motor, drive and control components from operationally detrimental high temperatures. The construction of each heater shall be built around a structurally reinforced, unitized housing and base made of G90 galvanized steel and shall be painted with a gray polyester powder paint finish. **Heaters** are to have a fully insulated cabinet with 1" thick, 1½ lb. density NFPA 90A thermal and acoustical insulation (mechanically fastened). The base of the heaters shall be fully, internally insulated with ½" thick, non-water absorbing, closed cell insulation. The control enclosure shall be directly accessible from the exterior of the heater and shall be watertight with a full perimeter gasket seal.

Blower: The blower shall be constructed with a discharge transition duct which is approximately 2½ wheel diameters in length to provide maximum regain of static pressure and uniform discharge air temperature. The blower shall be a double width double inlet (DWDI), forward-curved centrifugal fan. Heaters shall have permanently lubricated self-aligning sealed ball bearings, resiliently mounted for sound and vibration attenuation [SA350]. Fan bearings are to be located out of the heated air stream. Fan speed shall be at least 25% below the first critical speed for the shaft. The shaft shall be coated with a rust inhibitor.

Motor/Drive: The motor shall be a ❑¾ HP [SA250] ❑1 HP [SA350], ball bearing type, open drip-proof construction, designed for continuous duty at ❑115 volt [SA250] ❑208-230 volt [SA350] 1 phase 60 Hz, and shall have a 1.15 service factor. The motor bearings shall be located out of the heated air stream. The fan drive shall be ❑direct drive [SA250] ❑a heavy duty V-belt drive designed for a 1.5 minimum service factor based on motor horsepower[SA350].

Burner: The direct gas-fired burner shall be suitable for ❑natural gas ❑propane air. The burner shall have stainless steel burner plates with non-clogging orifices. The burner shall produce less than 5 PPM (parts per million) carbon monoxide and 0.5 PPM nitrogen dioxide over its entire firing range. **The burner shall be warranted for five years.** The burner shall be furnished with a low voltage/hot surface ignition system and stainless steel flame rod.

Gas Controls: The temperature control system shall be Maxitrol Series 44 modulation controls.

Controls: The heater shall be furnished with factory mounted controls to include low temperature cutout, entering air thermostat, pre-purge timer, patented low fire start control, service switches for blower and burner operation at the heater, IEC motor starter with ❑integral motor protection [SA250] ❑single phase overload protection [SA350], redundant gas valves, pressure regulator, electronic temperature controls and a low voltage, class 2 transformer for remote control thermostat wiring.

WARNING - DO NOT ATTEMPT TO SUBSTITUTE any direct gas-fired heater incapable of achieving 160°F discharge temperature. The technologies are different. Ability to heat the building will be jeopardized and energy costs will increase.

Factory Design Assistance

Take advantage of our 50+ years experience calculating heat loads and let us help design the most effective and energy efficient heating system for your building. Our proprietary Cal-Q-Heat® Program has become an industry standard for heat load calculations associated with warehouses, distribution centers, manufacturing plants, aircraft hangars, indoor recreational facilities, automobile service areas and other commercial/ industrial buildings.

Fill out the Cal-Q-Heat® form

- Contact your local Cambridge Sales Representative to obtain a Cal-Q-Heat® input data form. It will include a list of the required building parameters and operating conditions needed to provide a heat loss calculation. The minimum information required is shown at the right.
- The Cal-Q-Heat® form can also be downloaded from our website, filled out and faxed or emailed to Cambridge or your local representative.
- Remember, the heat load and heater design are only as good as the accuracy of the information provided.

Cambridge provides the following:

- Fast turnaround for heat loads and heater design information to meet your specific requirements.
- System design to include the number, size and location of Cambridge Blow-Thru® Space Heaters that will result in the most effective and energy efficient heating system for the building.
- Heater cut sheets in CAD format can be provided on request.

CAUTION:

All information provided by Cambridge is based on the use of energy efficient, Genuine Cambridge® direct gas-fired space heating equipment capable of achieving a discharge temperature of 160°F.

DO NOT ATTEMPT TO SUBSTITUTE any other type of indirect or direct gas-fired heating equipment. The technologies are different. The ability to heat a building will be jeopardized and energy costs will increase.

CAL-Q-HEAT® INFORMATION

Job Name:_____

Job Location:_____

Design Temperatures
Indoor: _____°F Outdoor:_____°F

Building Dimensions:
_____ft. W x _____ft. L x _____ft. H

Insulation
Roof R-value:_____ Walls R-value:_____

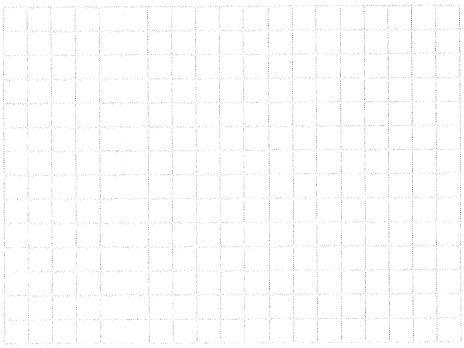
Gas Supply
❑ Natural Gas ❑ Propane Air

Exhaust
_____cfm ❑ Intermittent ❑ Continuous

Doors
Qty:_____ Size:____ft. W____ft. H ❑ Seals
Qty:_____ Size:____ft. W____ft. H ❑ Seals

❑ **LEED Project**

Provide building sketch with location of doors.



Model	Typical Heater Discharge Height1 (ft.)
SA250	10 - 15
SA350	10 - 20

NOTE: Consider accessibility and safety when selecting discharge height and mounting options.

[1]Distance from bottom of discharge to finished floor.

Typical Capacity Selection
For Standard Mounting Packages at sea level

Model	Motor HP	Gas Supply			Roof Top			Thru Wall			Under Roof		
		Inlet Size	Min. Pressure "WC	Max. Pressure "WC	Input MBH[3]	Inlet cfm	TESP[2] "WC	Input MBH[3]	Inlet cfm	TESP[2] "WC	Input MBH[3]	Inlet cfm	TESP[2] "WC
SA250	¾	½"	7	14	250	1,200	.06	250	1,200	.05	250	1,200	.07
SA350[6]	1	¾"	7	.14	350	1,620	.07	350	1,620	.07	350	1,620	.06

[2]Total External Static Pressure (TESP) is the total of all airflow resistances (in inches WC) from Mounting Package components. Any other system air flow resistances such as ductwork should be added to the above TESP figures.
[3]The MBH and cfm ratings are based upon an inlet temperature of -40°F.

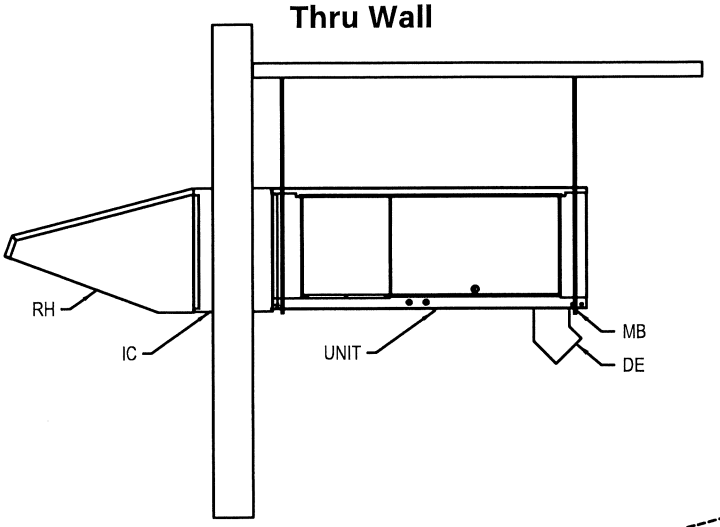
Heater Input Capacity: $Btu/hr = cfm \times \rho \times c_p \times 60 \times \Delta T \div 0.92$

Where:
cfm is the inlet air volume of the blower
ρ is the density[6] of the air handled by the blower (lb/ft^3)
$\mathbf{c_p}$ is the specific heat of the air (0.240 Btu/lb °F)
60 is the conversion from minutes to hours
ΔT is the temperature rise (160°F Max.)
0.92 is the conversion from sensible to total heat (output to input)

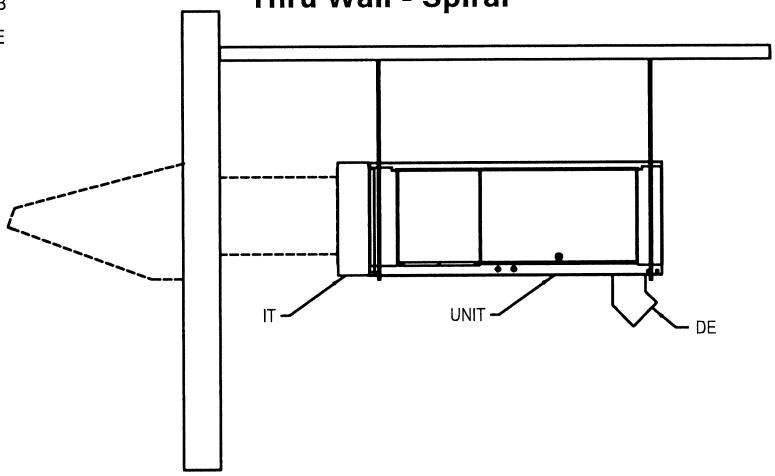
[4]Density is calculated from the following formula: $r = 1.32605 \times$ (barometric pressure ÷ °R)
Where: barometric pressure is in terms of inches of mercury ("Hg) and °R = (460 + T_{Inlet} °F)
[5]Positive shut-off pressure regulators are required when gas supply pressure exceeds 14" WC
[6]The SA350 was discontinued in 2022.

Standard Mounting Packages

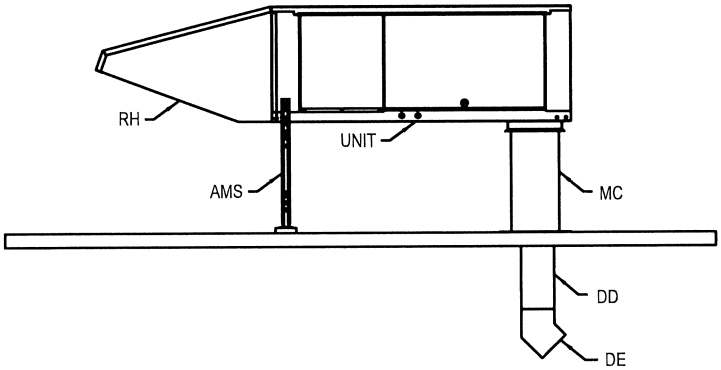
Thru Wall



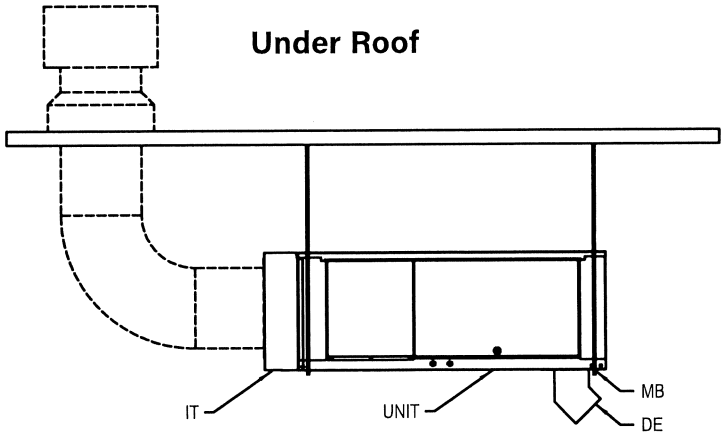
Thru Wall - Spiral



Roof Top



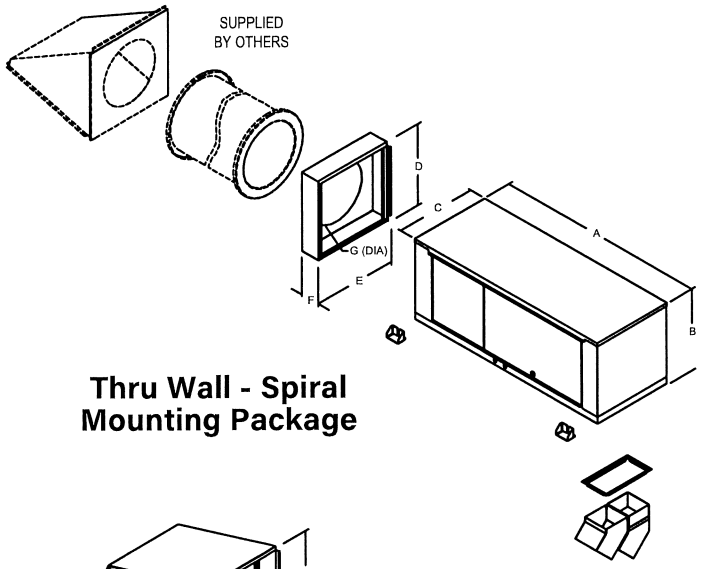
Under Roof



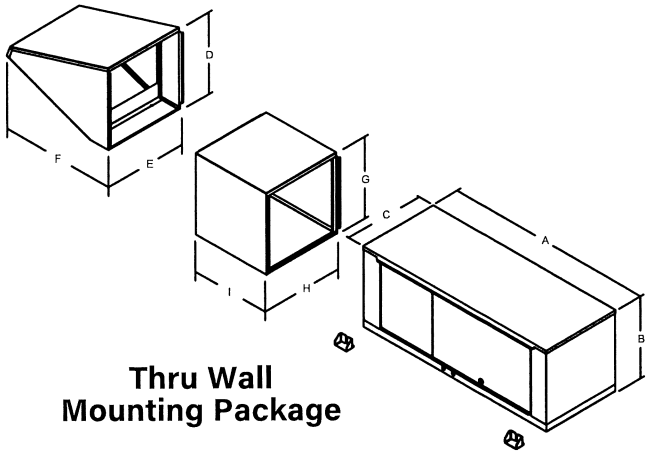
DD	**Discharge Duct:** The 50" discharge duct for roof top mounting package extends through the mounting curb.
DE	**Directional Elbows**[1]**:** Used to optimize heat distribution by directing air to meet the requirements of specific field applications. Field mounting is required.
EAT	**Entering Air Thermostat**[1]**:** The EAT automatically turns off the burner when the outdoor temperature approaches the EAT setpoint temperature (45°F to 70°F) while maintaining blower operation for ventilation.
EFC	**Exhaust Fan Contact:** The EFC is an auxiliary dry contact mechanically interlocked to the operation of the motor starter of the heater. The dry contact is typically wired into an exhaust fan control circuit to activate an exhaust fan.
EDL	**EDL - Space Heating Gas Trains**[1]**:** An electronic discharge air temperature control which utilizes a discharge temperature sensor in conjuction with an electronic proportioning gas valve for maintaining the selected discharge air temperature. This control will permit manual adjustment of the discharge air temperature from 110°F to 160°F via the selector on the amplifier located in the heater's electrical control enclosure.
IC	**Inlet Collar:** Insulated collar to connect rain hood to unit. Can accommodate up to 21" thick walls.
IT	**Inlet Transition:** Insulated collar to connect spiral duct to unit.
LFS	**Low Fire Start**[1]**:** The LFS function limits the initial heater firing for the first fifteen seconds of a heating cycle.
LTC	**Low Temperature Cutout with Alarm**[1]**:** The LTC alarm circuit functions to shut down the blower in approximately four (4) minutes if either of the following occurs: (1) The inlet temperature drops below the LTC setpoint (40, 45, 50 or 55°F) in the Ventilation mode; or (2) The gas valve fails to remain energized during a heating cycle.
MB	**Mounting Brackets:** Provided for installation from hanging rods.
MC	**Mounting Curb:** The insulated 24" mounting curb supports the discharge end of the unit. This mounting curb is designed for conventional, flat roof applications. Counterflashing is included. A cant, if required, is provided by others.
AMS	**Mounting Stand:** An adjustable mounting stand (18¾" to 26¼") is used as the rear heater support on the roof top mounting package. A cant, if required, is provided by others. A 1½" thick piece of pressure treated wood is required (not provided) for the base of the stand to rest on to meet the overall height of 24".
PT	**Pre-Purge Timer**[1]**:** The PT function provides a minimum of four air changes in the heater housing and any attached inlet accessories or field installed ductwork prior to an ignition attempt. The time is selectable at 2, 4, 8, 16, or 32 seconds.
RH	**Rain Hood:** Provided with a metal inlet screen.
SS	**Service Switches**[1]**:** Located in the heater to allow local control by Service Technician when servicing the heater.

[1] Included with all units.

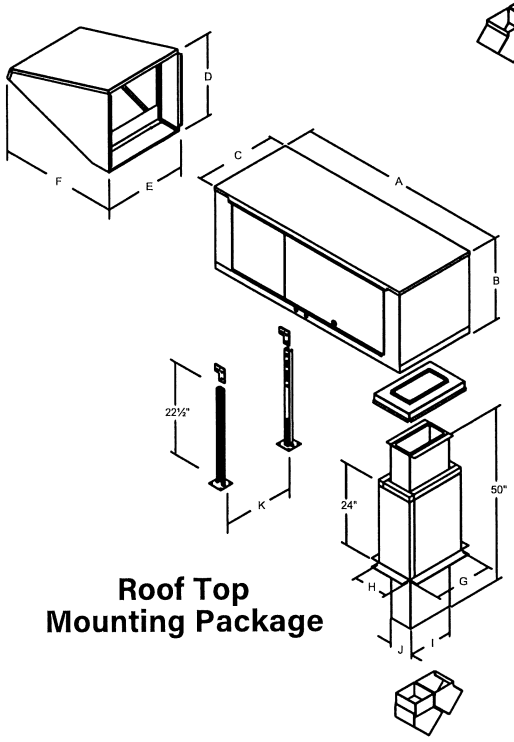
Unit Weights and Dimensions



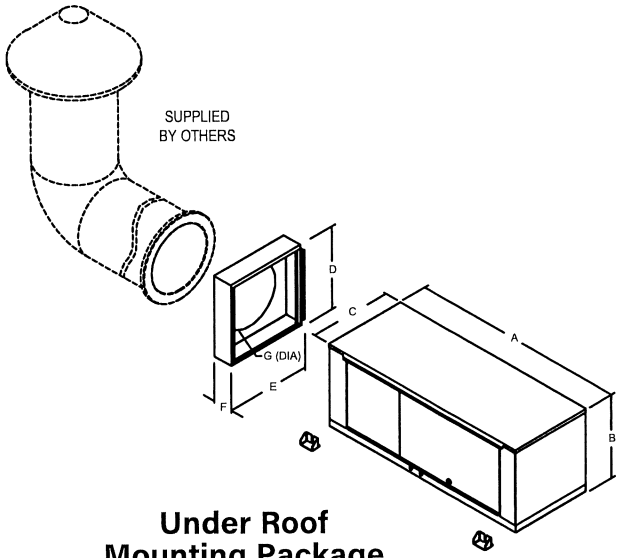
Thru Wall - Spiral Mounting Package



Thru Wall Mounting Package

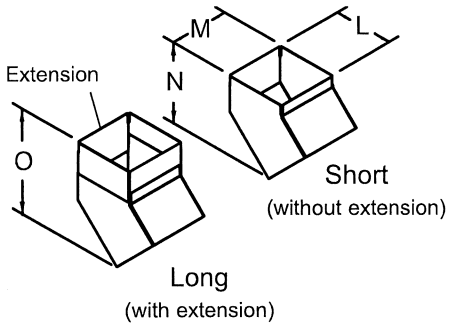


Roof Top Mounting Package

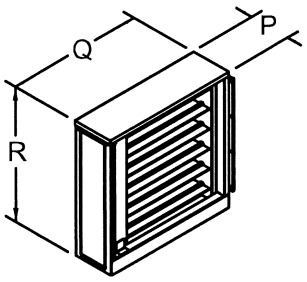


Under Roof Mounting Package

Model	Mounting Package	Dimensions											Weight (lbs.)
		A	B	C	D	E	F	G	H	I	J	K	
SA250	Roof Top	55"	24¼"	21"	21"	21"	30½"	17¼"	10"	13¼"	6½"	24½"	265
	Thru Wall							21"	21"	21"			230
	Thru Wall - Spiral						5"	16"					200
	Under Roof												
SA350	Roof Top	73"	25½"	25½"	24"	23½"	36¾"	26½"	14¾"	22¼"	11"	25½"	585
	Thru Wall							24"	25½"	21"			505
	Thru Wall - Spiral						5"	19"					445
	Under Roof												



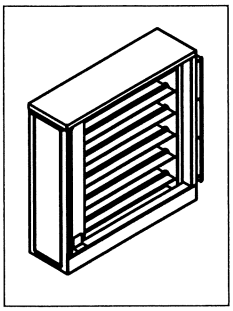
Directional Elbows



Inlet Damper (optional)

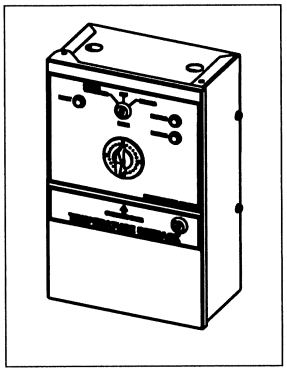
Model	Directional Elbows				Inlet Damper			
	Dimensions				Dimensions			Weight (lbs.)
	L	M	N	O	P	Q	R	
SA250	6½"	6½"	9¾"	12½"	(Internal)			10
SA350	11¼"	11¼"	13¾"	18¼"	7¾"	25½"	25½"	45

S-Series Selection



Motorized Inlet Damper (MID)
A two-position damper assembly that closes when the blower is not operating.

Electrical Control Options



CE-Smart System

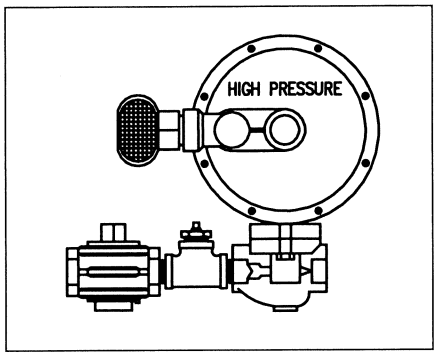
A lockable NEMA 1 enclosure (9"w x 14"h x 5"d) equipped with a three position keylock selector switch (Summer Ventilation - Off - Heating), a combination operating electronic thermostat and seven day programmable time clock (TSS), an override timer and indicating lights for blower operation (green), burner operation (red) and reset (amber). The TSS accommodates separate programming for summer ventilation and heating modes.

The operator can control the heater from this panel provided the heater mounted disconnect is in the "ON" position and the service switches are in the "REMOTE" position. The TSS utilizes an encapsulated thermistor to sense space temperature. The adjustable differential is set by the selection of the ON and OFF temperatures. This control can accommodate separately programmed schedules for both the ventilation and heating modes.

An intermittent/continuous control allows the heater to be operated continuously at modulating discharge temperature for make-up air (EDSM - Continuous Make-Up Air Mode) during occupied hours and operated intermittently based on a thermostat at maximum discharge temperature (EDL - Space Heating Mode) for space heating during unoccupied hours.

Terminals are provided for wiring of an exhaust fan interlock (EFI) contacts provided by others. Typically used when the heater shall come on when an exhaust fan comes on.

High Pressure Regulator (HPR)
A positive shut-off high pressure regulator is required when the gas supply pressure exceeds the maximum gas pressure requirements or is specified by local codes or utilities. It is sized according to the gas supply pressure and the capacity requirements of the heater. It must be vented to the outdoors. (Specify gas supply pressure.)



Summary of Limited Warranty

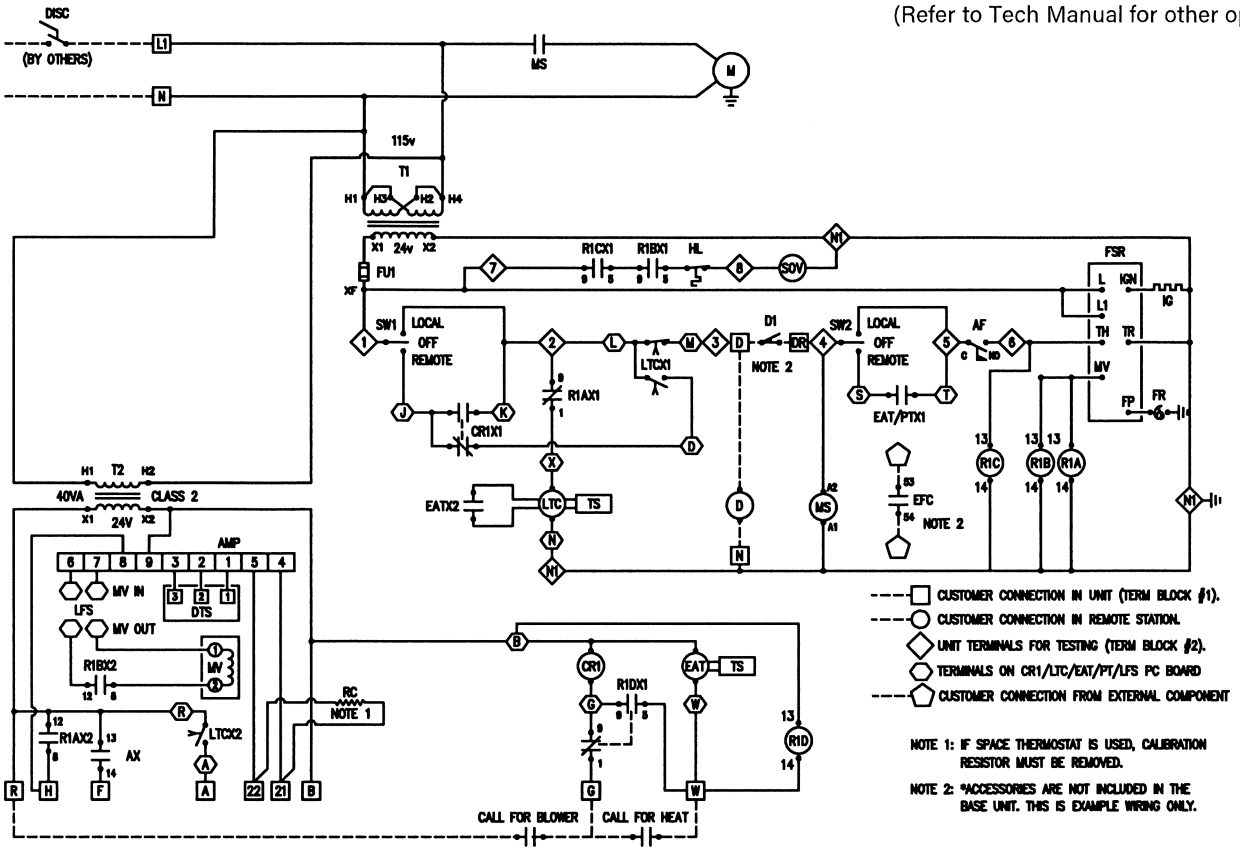
Cambridge Air Solutions warrants all SA-Series products, including all components and sub-components thereof, to be free from defects in material and workmanship for a period of **twenty-four (24) months** from date of shipment, with the exception of the burner which is warranted for **five (5) years**, provided the product is properly installed and operated under normal conditions in accordance with the Cambridge Technical Manual and any other applicable instructions and in conformance with national and local codes.

For complete warranty, see Standard Terms and Conditions in the product catalog or the Technical Manual.

Wiring Diagram

For SA250 only

(Refer to Tech Manual for other options)



Symbol	Description
AF	Air Flow Switch
AMP	Amplifier Solid State
AX	Auxiliary Contact
CR1	Control Relay
D & D1	Damper Motor & End Switch
DTS	Discharge Temperature Sensor
EAT	Entering Air Thermostat
EFC	Exhaust Fan Contact
FR	Flame Rod
FSR	Flame Safeguard Relay (HSI)

Symbol	Description
FU1	Fuse 24 Volt Control
HL	High Limit
IG	Igniter
LFS	Low Fire Start
LTC	Low Temperature Cutout
M	Motor
MS	Motor Starter
MV	Modulating Valve
PCB	Printed Circuit Board
PT	Purge Timer

Symbol	Description
R1A,B,C	Relays - Gas Valve
RID	Relay - Thermostat Control
RC	Resistor - Calibration
SOV	Shut-Off Valve - Gas
SW1	Service Switch - Fan
SW2	Service Switch - Heat
T1	Multi-Tap Transformer (24 & 120 Volt)
T2	Class 2 Transformer (24 Volt)
TS	Temperature Sensor (LTC/EAT)

Model	Motor HP	Heater Amperage Requirements		
		115V/1Ph	208V/1Ph	230V/1Ph
SA250	¾	12.3		
SA350	1		9.8	9.3